SRM IST

Delhi - NCR

B. Tech CSE III / 6th sem
Assignment - 4

Subject: Data Science

Subject Code: 21CSS303T

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Explain the concept of data wrangling & its importance in data analysis.

Data wrangling is the process of collecting, cleaning, transforming & organizing raw data into a usable format for analysis.

Steps involved in data wrangling

- Data Collection
- Data Cleaning
- Data Transformation
- Data Integration
- Data Validation

Importance in data analysis

- Improves data quality
- Increases accuracy of analysis
- Saves time in later stages
- Helps in better decision-making

Example: Suppose a sale dataset contains:

- missing product prices
- duplicate customer records
- inconsistent data formats

Explain string manipulation in python with suitable code examples.

String manipulation means performing operations on text data such as splitting, replacing, searching, formatting strings.

Common String operations

Converting Case

```
text = "Python Programming"
print (text.upper())
print (text.lower())
print (text.title())
```

b) Finding length of string

```
text = "Data Science"
print (len (text))
```

c) Replacing text

```
text = "I like Python"
new_text = text.replace ("Python", "Java")
print (new_text)
```

d) Splitting String

```
text = "apple, banana, mango"
fruits = text.split (",")
print (fruits)
```

e) Joining string

```
words = ['Data', 'Science', 'Python']
result = " ".join (words)
print (result)
```

f) Checking substring

```
text = "machine learning"
print ("learning" in text)
```

stance:

ng manipulation is widely used in:
text preprocessing
natural language processing
data cleaning
extracting useful information

ain the general techniques & programming prac
ed for efficient handling of large datasets in
ata science.

Modelling large datasets efficiently is very importa
in data science because datasets may contain mil
of rows.

General techniques
a) Use efficient libraries
Use libraries optimized for large data:
- pandas - numpy - pyspark

```
import pandas as pd
df = pd.read_csv("data.csv")
```

b) Load data in chunks
Instead of loading the whole file at once, loa
in parts for chunks in pd.read_csv("large_file.
chara tter size = 1000):
print (chunk.head())
This reduces memory usage.

c) Use appropriate data types
df['age'] = df['age'].astype(int32)
Smaller data types save memory

are unnecessary column

df = df [['name', 'salary']]

Keeps only useful features

Use vectorized operations

Avoids loops where possible

df ['bonus'] = df ['salary'] * 0.10

Vectorized operation are faster than for loops.

Explain different data transformation techniques used in preprocessing.

Data transformation in the process of converting raw-data into a suitable format for analysis & machine learning.

Common transformation tech.

a) Normalization

Scales values between 0 & 1

Formula: $x' = \frac{x - x_{min}}{x_{max} - x_{min}}$

b) Standardization

Transformation data to mean = 0 & standard deviation = 1

$z = \frac{x - \mu}{\sigma}$

c) Encoding categorical data

Converting text categories into numeric values.

From sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df ['gender'] = le.fit_transform(df ['gender'])

g transformation

used for skewed data

import numpy as np

df ['income'] = np.log(df ['income'])

e) Binning

Converts continous values into graphs

df [age-graph] = pd.cut(df ['age'], bins=[0,18,40,60,100])

f) Feature scaling

Make all features comfortable

used in:

- KNN - SVM - Linear Regression

Importance

- Improves model performance
- reduce bias
- handles skewness.